

# Do Option Characteristics Predict Underlying Stock Returns in the Cross-Section?

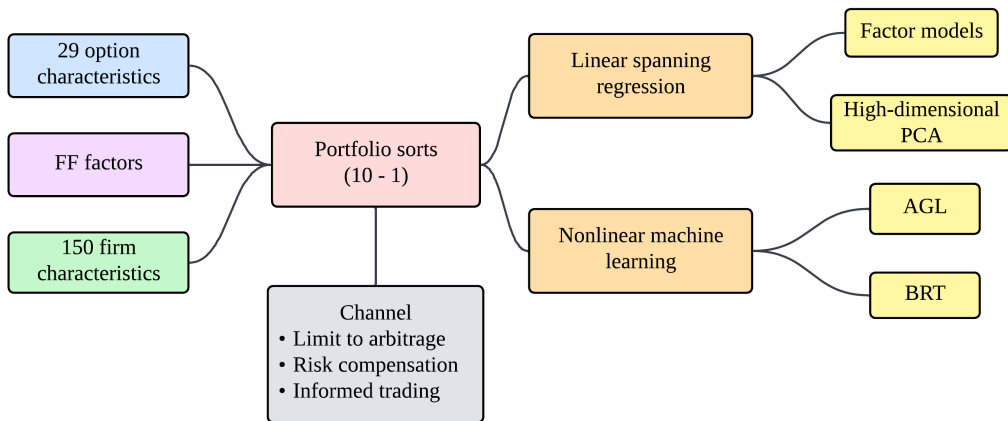
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(2026, *MS*)

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# Overview



# Motivation

- Limitation of existing cross-sectional asset pricing.
  - Virtually all studies (Feng et al., 2021; Kelly and Xiu, 2023) exclusively rely on observable firm characteristics and past realization of return and volume.
  - Ignoring forward-looking information from option markets which may not be captured by observable characteristics.
- Potential information for cross-section of stock return.
  - Widely recognized that option market contain rich information about underlying asset, economic state, associated risk (Ross, 2015; Andersen et al., 2015).
  - Option-implied characteristics help explain cross-section of stock return (Bali, 2009; Cremers, 2010; An, 2014; Ge, 2016), but only control for very small set of factors.
  - Examining information embedded in option jointly with large number of known cross-sectional factors remain undertaken.

# Research Question

- Do option markets provide incremental forward information about stock return?
  - Yes, option prices contain incremental information about future.
- Which option characteristics matter?
  - Only 6 option characteristics related to implied volatility (IV) spread matter after controlling large set of firm characteristics.
- Why do they matter?
  - H1: risk compensation
  - H2: limit to arbitrage
  - H3: informed trading

## Contribution

- Contribute to literature on cross-sectional return predictability.
  - Prior: large number of new factors can predict return in cross-section (Cochrane, 2011).
  - Limit: mainly rely on observable firm characteristics and past realization of return and volume (Feng et al., 2020; Freyberger et al., 2020; Gu et al., 2020).
  - This paper: incorporate comprehensive set of option characteristics to examine whether option market provide additional forward-looking information.
- Contribute to literature on option-based asset pricing.
  - Prior: significant stock return predictability from option characteristics (Bali and Hovakimian, 2009; Cremers and Weinbaum, 2010; An et al., 2014; Ge et al., 2016).
  - Limit: typically rely on limited set of control variables.
  - This paper: control large set of firm factors and find only few option characteristics survive comprehensive controls.

# Data

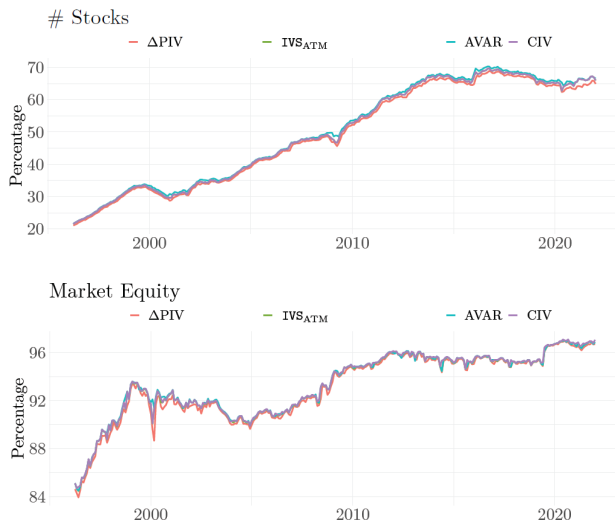
- Option market data
  - Implied volatility surfaces, option and strike price, trading volume from OptionMetrics.
  - Monthly dataset from 1996 to 2021.
  - 29 option characteristics from literature on option-based cross-section asset pricing.
  - Focus on publicly availability and straightforward variable construction.

|                                | Volume Factors |            | IV Factors |     |              |              |             |             |                    |      | Price Factors |      |         |         |         | Other Factors |              |                |                |                |             |             |                      |     |
|--------------------------------|----------------|------------|------------|-----|--------------|--------------|-------------|-------------|--------------------|------|---------------|------|---------|---------|---------|---------------|--------------|----------------|----------------|----------------|-------------|-------------|----------------------|-----|
|                                | O/S            | Signed O/S | CIV        | PIV | $\Delta CIV$ | $\Delta PIV$ | $IVS_{ATM}$ | $IVS_{OTM}$ | $\Delta IVS_{OTM}$ | Skew | IV PCA        | AVAR | $VAR^+$ | $VAR^-$ | $VAR^Q$ | KT            | Vol. of vol. | $\beta_{Jump}$ | $\beta_{Vol.}$ | $\beta_{Skew}$ | co-skewness | co-kurtosis | Option-implied betas | VRP |
| Johnson and So (2012)          | X              |            |            |     |              |              |             |             |                    |      |               |      |         |         |         |               |              |                |                |                |             |             |                      |     |
| Ge et al. (2016)               |                | X          |            |     |              |              |             |             |                    |      |               |      |         |         |         |               |              |                |                |                |             |             |                      |     |
| An et al. (2014)               |                |            |            |     | X            | X            |             |             |                    | X    |               |      |         |         |         |               |              |                |                |                |             |             |                      | X   |
| Bakshi and Kapadia (2003)      |                |            |            |     |              |              |             |             |                    |      |               |      |         |         |         |               |              |                |                |                |             |             |                      | X   |
| Bali and Hovakimian (2009)     |                |            | X          | X   |              |              | X           |             |                    |      |               |      |         |         |         |               |              |                |                |                |             |             |                      | X   |
| Kilic and Shaliastovich (2019) |                |            |            |     |              |              |             |             |                    |      |               |      | X       | X       |         |               |              |                |                |                |             |             |                      | X   |
| Carr and Wu (2009)             |                |            |            |     |              |              |             |             |                    |      |               |      |         |         |         |               |              |                |                |                |             |             |                      | X   |
| Xing et al. (2010)             |                |            |            |     |              |              |             | X           |                    |      |               |      |         |         |         |               |              |                |                |                |             |             |                      | X   |
| Yan (2011)                     |                |            |            |     |              |              | X           | X           |                    |      |               |      |         |         |         |               |              |                |                |                |             |             |                      |     |
| Conrad et al. (2013)           |                |            |            |     |              |              |             |             |                    | X    |               |      |         |         |         | X             |              |                |                | X              |             |             |                      |     |
| Stilger et al. (2017)          |                |            |            |     |              |              |             |             |                    | X    |               |      |         |         |         |               |              |                |                | X              |             |             |                      |     |
| Cremers and Weinbaum (2010)    |                |            |            |     |              |              | X           |             |                    | X    |               |      |         |         |         |               |              |                |                |                |             |             |                      |     |
| Cremers et al. (2015)          |                |            |            |     |              |              |             | X           |                    | X    |               |      |         |         |         |               |              | X              | X              |                |             |             |                      | X   |
| Kadan and Tang (2020)          |                |            |            |     |              |              |             |             |                    |      |               |      |         |         |         | X             |              |                |                |                |             |             |                      |     |
| Tang (2019)                    |                |            |            |     |              |              |             |             |                    |      |               | X    | X       | X       | X       |               |              |                |                |                |             |             |                      |     |
| Martin and Wagner (2019)       |                |            |            |     |              |              |             |             |                    |      |               |      |         |         |         |               |              |                |                |                |             |             |                      |     |
| Baltussen et al. (2018)        |                |            |            |     |              |              |             |             |                    |      |               |      |         |         |         |               | X            |                |                |                |             |             |                      |     |
| Christoffersen et al. (2021)   |                |            |            |     |              |              |             |             |                    |      |               |      |         |         |         |               |              |                |                |                | X           | X           |                      |     |
| Boloorforoosh et al. (2020)    |                |            |            |     |              |              |             |             |                    |      |               |      |         |         |         |               |              |                |                |                |             |             | X                    |     |
| Buss and Vilkov (2012)         |                |            |            |     |              |              |             |             |                    |      |               |      |         |         |         |               |              |                |                |                |             |             | X                    |     |
| Muravyev et al. (2020)         |                |            |            |     |              |              |             |             |                    |      | X             |      |         |         |         |               |              |                |                |                |             |             |                      |     |

# Data

- Firm fundamental data
  - Firm-level accounting and market characteristics from Compustat.
  - 150 firm characteristics from Jensen et al.(2023) as comprehensive control variables.
  - Missing value imputed following Freyberger et al.(2025).
- Stock market data
  - Stock price, trading volume, return from CRSP.
  - Construct stock return outcomes and portfolio sorting strategy.
- Equity lending cost data
  - Daily cost of borrowing score from Markit.
  - Examine whether limit to arbitrage explain predictability.

# Option Coverage



# Empirical Evaluation of Option Characteristics

- Portfolio sorting
  - Sort stocks into decile portfolios based on each option characteristic each month.
  - Construct 10-1 portfolios with equal-weighted average return.
- Linear spanning regression
  - Alpha test using large set of control model and variable: CAPM, FF3, FF4, FF5, FF6 (FF5+MOM), FF6+Liquidity, FF6+Volatility, HXZ, HXZM, PC5, PC7.
  - PC5 and PC7 take 5 and 7 principal components of 150-characteristic-sorted portfolios.
- Nonlinear machine learning of individual stock return
  - AGL (Adaptive Group LASSO): option characteristics jointly with large set of firm characteristics; retain characteristic only when its Legendre polynomials provide incremental predictive information.
  - BRT (LightGBM): each input 1 characteristic to be tested and 150 firm controls; rank importance by its relative influence score.

# Performance Statistics on High-Low Portfolios

|                    | Mean (%) | Standard deviation (%) | Sharpe ratio | Low Pf. (%) | High Pf. (%) | Skewness | Kurtosis | Maximum drawdown |
|--------------------|----------|------------------------|--------------|-------------|--------------|----------|----------|------------------|
| CIV                | -5.10    | 37.59                  | -0.14        | 8.47        | 3.37         | 0.54     | 3.27     | 84.88            |
| PIV                | -8.81    | 37.29                  | -0.24        | 9.13        | 0.32         | 0.59     | 3.39     | 76.30            |
| $\Delta$ CIV       | 4.86     | 12.24                  | 0.40         | 3.81        | 8.67         | 0.58     | 12.49    | 35.94            |
| $\Delta$ PIV       | -2.38    | 12.09                  | -0.20        | 7.51        | 5.13         | -0.20    | 17.18    | 52.51            |
| O/S                | -6.97    | 15.46                  | -0.45        | 11.51       | 4.54         | 0.70     | 10.06    | 61.18            |
| VRP                | -0.16    | 13.00                  | -0.01        | 6.15        | 5.99         | -0.59    | 3.60     | 40.62            |
| $IVS_{ATM}$        | -11.95   | 10.73                  | -1.11        | 11.61       | -0.35        | -2.11    | 25.32    | 26.82            |
| $IVS_{OTM}$        | -8.53    | 12.57                  | -0.68        | 12.14       | 3.61         | -1.97    | 16.75    | 34.42            |
| $\Delta IVS_{ATM}$ | -6.77    | 7.53                   | -0.90        | 10.15       | 3.38         | 0.07     | 5.12     | 17.50            |
| $\Delta IVS_{OTM}$ | -5.63    | 6.82                   | -0.83        | 10.38       | 4.75         | -1.18    | 6.71     | 13.16            |
| Skew               | -7.95    | 10.19                  | -0.78        | 11.43       | 3.48         | -0.89    | 5.34     | 19.20            |
| $\beta_{Jump}$     | -0.82    | 15.51                  | -0.05        | 5.48        | 4.66         | 0.28     | 2.98     | 64.15            |
| $\beta_{Vol.}$     | -0.02    | 13.43                  | 0.00         | 5.48        | 5.47         | -1.24    | 5.84     | 71.76            |
| $\beta_{VIX}$      | -6.29    | 17.79                  | -0.35        | 8.53        | 2.24         | -1.76    | 12.03    | 42.29            |
| $\beta_{Skew}$     | -7.38    | 16.67                  | -0.44        | 9.26        | 1.88         | -0.49    | 5.66     | 51.76            |
| $\beta_{VRP}$      | -0.65    | 18.53                  | -0.04        | 6.41        | 5.76         | -0.51    | 7.96     | 73.85            |
| $\beta_{IVSOTM}$   | -5.54    | 16.56                  | -0.33        | 8.23        | 2.69         | -0.40    | 5.02     | 62.09            |
| $\beta_{TSCALL}$   | -3.53    | 16.96                  | -0.21        | 7.19        | 3.67         | -1.44    | 8.10     | 45.87            |
| $\beta_{TSPUT}$    | -4.12    | 16.51                  | -0.25        | 7.74        | 3.63         | -1.54    | 10.34    | 51.10            |
| AVAR               | -10.03   | 13.93                  | -0.72        | 10.83       | 0.81         | -0.16    | 7.49     | 41.68            |
| VAR <sup>+</sup>   | -5.51    | 36.60                  | -0.15        | 8.72        | 3.21         | 0.60     | 4.26     | 81.31            |
| VAR <sup>-</sup>   | -8.98    | 37.15                  | -0.24        | 9.81        | 0.83         | 0.48     | 3.42     | 78.23            |
| VAR <sup>Q</sup>   | -7.22    | 37.25                  | -0.19        | 9.36        | 2.14         | 0.50     | 3.61     | 77.87            |
| KT                 | -6.70    | 37.31                  | -0.18        | 9.35        | 2.65         | 0.55     | 3.82     | 80.55            |
| Vol. of vol.       | -5.25    | 9.47                   | -0.55        | 11.15       | 5.91         | 0.06     | 3.55     | 24.17            |
| TSCALL             | -5.62    | 14.83                  | -0.38        | 8.61        | 2.99         | -0.19    | 12.07    | 21.91            |
| TSPUT              | -5.25    | 14.91                  | -0.35        | 7.92        | 2.66         | 0.18     | 10.08    | 29.90            |
| $\Delta$ TSCALL    | 0.64     | 9.78                   | 0.07         | 6.33        | 6.97         | -2.02    | 18.31    | 34.59            |
| $\Delta$ TSPUT     | -2.73    | 9.88                   | -0.28        | 7.61        | 4.87         | -1.83    | 18.47    | 43.46            |

# Alpha under Factor Models

|                         | CAPM     | FF3      | FF4      | FF5      | FF7a     | FF7b     | HXZ      | HXZM     | PC5      | PC7      |
|-------------------------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|
| CIV                     | -1.54*** | -1.49*** | -1.03*** | -0.51    | -0.07    | -0.26    | -0.24    | -0.22    | -0.44*   | 0.49**   |
| PIV                     | -1.83*** | -1.78*** | -1.33*** | -0.79**  | -0.36    | -0.54**  | -0.53    | -0.51    | -0.78*** | 0.11     |
| ΔCIV                    | 0.27     | 0.27     | 0.32     | 0.33*    | 0.35*    | 0.28*    | 0.34     | 0.35     | 0.47**   | 0.24     |
| ΔPIV                    | -0.30*   | -0.30*   | -0.23    | -0.25    | -0.21    | -0.26    | -0.21    | -0.19    | -0.11    | -0.40    |
| O/S                     | -0.77*** | -0.68*** | -0.84*** | -0.43*** | -0.51*** | -0.54*** | -0.50*   | -0.56*** | -0.28*   | 0.08     |
| VRP                     | 0.05     | 0.05     | 0.03     | 0.07     | 0.05     | 0.02     | 0.12     | 0.11     | 0.11     | -0.05    |
| IVS <sub>ATM</sub>      | -0.92*** | -0.94*** | -1.08*** | -1.08*** | -1.18*** | -1.15*** | -1.19*** | -1.21*** | -1.43*** | -1.10*** |
| IVS <sub>OTM</sub>      | -0.51**  | -0.54*** | -0.69*** | -0.75*** | -0.88*** | -0.81*** | -0.96*** | -0.96*** | -0.94*** | -0.96*** |
| ΔIVS <sub>ATM</sub>     | -0.52*** | -0.52*** | -0.51*** | -0.64*** | -0.63*** | -0.60*** | -0.56*** | -0.56*** | -0.63*** | -0.62*** |
| ΔIVS <sub>OTM</sub>     | -0.43*** | -0.43*** | -0.48*** | -0.50*** | -0.53*** | -0.51*** | -0.49*** | -0.50*** | -0.53*** | -0.51*** |
| Skew                    | -0.53*** | -0.54*** | -0.63*** | -0.66*** | -0.75*** | -0.67*** | -0.82*** | -0.82*** | -0.77*** | -0.83*** |
| $\beta_{\text{Jump}}$   | -0.29    | -0.29    | -0.14    | -0.01    | 0.12     | 0.07     | 0.10     | 0.11     | -0.23    | 0.24     |
| $\beta_{\text{Vol}}$    | 0.09     | 0.12     | 0.03     | 0.18     | 0.12     | 0.12     | 0.06     | 0.04     | 0.41*    | 0.01     |
| $\beta_{\text{VIX}}$    | -0.42    | -0.43*   | -0.54**  | -0.65*** | -0.73*** | -0.77*** | -0.77*** | -0.77*** | -0.43    | -0.34    |
| $\beta_{\text{Skew}}$   | -0.56*   | -0.54*   | -0.47    | -0.48    | -0.43    | -0.48    | -0.47    | -0.46    | -0.24    | -0.15    |
| $\beta_{\text{VRP}}$    | 0.27     | 0.26     | 0.07     | -0.01    | -0.20    | -0.17    | -0.16    | -0.18    | -0.01    | -0.34    |
| $\beta_{\text{IVSOTM}}$ | -0.42    | -0.41    | -0.31    | -0.36    | -0.29    | -0.35    | -0.35    | -0.34    | -0.11    | 0.01     |
| $\beta_{\text{TSCALL}}$ | -0.20    | -0.21    | -0.28    | -0.43*   | -0.48*   | -0.52*   | -0.55**  | -0.55**  | -0.17    | -0.10    |
| $\beta_{\text{TSPUT}}$  | -0.26    | -0.27    | -0.28    | -0.45*   | -0.45    | -0.50*   | -0.51*   | -0.50*   | -0.23    | -0.12    |
| AVAR                    | -1.10*** | -1.06*** | -1.00*** | -0.66*** | -0.60*** | -0.59*** | -0.72*** | -0.73*** | -0.74*** | -0.35*   |
| VAR <sup>+</sup>        | -1.50*** | -1.46*** | -1.00*** | -0.45    | -0.03    | -0.20    | -0.21    | -0.18    | -0.43**  | 0.30*    |
| VAR <sup>-</sup>        | -1.82*** | -1.77*** | -1.34*** | -0.77**  | -0.36    | -0.53**  | -0.52    | -0.51    | -0.78*** | 0.14     |
| VAR <sup>Q</sup>        | -1.68*** | -1.63*** | -1.18*** | -0.63**  | -0.21    | -0.39    | -0.39    | -0.37    | -0.63*** | 0.21     |
| KT                      | -1.64*** | -1.59*** | -1.13*** | -0.59*   | -0.16    | -0.34    | -0.33    | -0.31    | -0.58**  | 0.26     |
| Vol. of. vol.           | -0.40**  | -0.42**  | -0.31*   | -0.28    | -0.23    | -0.23    | -0.32*   | -0.30*   | -0.32*   | -0.46**  |
| TSCALL                  | -0.71*** | -0.67*** | -0.50**  | -0.27    | -0.14    | -0.19    | -0.28    | -0.26    | -0.02    | 0.18     |
| TSPUT                   | -0.71*** | -0.67*** | -0.49*** | -0.31*   | -0.16    | -0.21    | -0.31*   | -0.29    | -0.05    | 0.00     |
| ΔTSCALL                 | 0.00     | 0.00     | 0.03     | 0.06     | 0.07     | 0.05     | 0.03     | 0.04     | 0.05     | -0.11    |
| ΔTSPUT                  | -0.26    | -0.25    | -0.25    | -0.27    | -0.27    | -0.28    | -0.25    | -0.25    | -0.21    | -0.44*   |

# Selected Predictors in Nonparametric Models

|                    | N       | Sample period     | BRT     | AGL |
|--------------------|---------|-------------------|---------|-----|
| AVAR               | 755,277 | Feb 1996–Dec 2021 | 10/151  | Y   |
| $\beta_{IVSOTM}$   | 755,982 | Feb 1996–Dec 2021 | 17/151  | Y   |
| $\beta_{Skew}$     | 755,982 | Feb 1996–Dec 2021 | 14/151  | Y   |
| $\beta_{TSCALL}$   | 755,982 | Feb 1996–Dec 2021 | 34/151  | Y   |
| $\beta_{TSPUT}$    | 755,982 | Feb 1996–Dec 2021 | 16/151  | Y   |
| $\beta_{VIX}$      | 755,982 | Feb 1996–Dec 2021 | 5/151   | Y   |
| $\beta_{VRP}$      | 755,982 | Feb 1996–Dec 2021 | 90/151  | N   |
| $\beta_{Jump}$     | 748,502 | May 1996–Dec 2021 | 110/151 | N   |
| $\beta_{Vol}$      | 748,502 | May 1996–Dec 2021 | 64/151  | N   |
| CIV                | 756,589 | Feb 1996–Dec 2021 | 8/151   | N   |
| $\Delta IVS_{ATM}$ | 744,016 | Mar 1996–Dec 2021 | 34/151  | Y   |
| $\Delta IVS_{OTM}$ | 744,016 | Mar 1996–Dec 2021 | 44/151  | Y   |
| $\Delta TSCALL$    | 747,773 | Mar 1996–Dec 2021 | 85/151  | N   |
| $\Delta TSPUT$     | 747,773 | Mar 1996–Dec 2021 | 31/151  | Y   |

|               |         |                   |        |   |
|---------------|---------|-------------------|--------|---|
| $\Delta CIV$  | 743,725 | Mar 1996–Dec 2021 | 30/151 | Y |
| $\Delta PIV$  | 743,725 | Mar 1996–Dec 2021 | 27/151 | N |
| $IVS_{ATM}$   | 755,990 | Feb 1996–Dec 2021 | 4/151  | Y |
| $IVS_{OTM}$   | 755,990 | Feb 1996–Dec 2021 | 11/151 | Y |
| KT            | 755,277 | Feb 1996–Dec 2021 | 9/151  | N |
| $VAR^-$       | 755,277 | Feb 1996–Dec 2021 | 5/151  | Y |
| O/S           | 713,187 | Feb 1996–Dec 2021 | 7/151  | Y |
| PIV           | 756,589 | Feb 1996–Dec 2021 | 1/151  | Y |
| $VAR^+$       | 755,277 | Feb 1996–Dec 2021 | 10/151 | N |
| $VAR^Q$       | 755,277 | Feb 1996–Dec 2021 | 9/151  | N |
| Skew          | 756,589 | Feb 1996–Dec 2021 | 12/151 | Y |
| TSCALL        | 755,992 | Feb 1996–Dec 2021 | 8/151  | Y |
| TSPUT         | 755,992 | Feb 1996–Dec 2021 | 8/151  | Y |
| Vol. of. vol. | 756,146 | Feb 1996–Dec 2021 | 20/151 | N |
| VRP           | 755,277 | Feb 1996–Dec 2021 | 69/151 | N |

# Information, Limits to Arbitrage, or Risk Compensation?

- Source of predictive power of option characteristics associated with IV spread
  - **Limit-to-arbitrage:** Excess return for IV spread subsumed when conditioning on lending costs data (Yan, 2011).
  - **Risk-return:** Excess return for IV spread positive relate with risk measures and returns (Muravyev et al., 2022).
  - **Informed-trading:** Excess return mainly driven by short (or long) leg indicate incremental information among short (or long) sellers (Cremers and Weinbaum, 2010).

# Characteristics for Portfolios Sorted by Option Characteristics

|                             | Portfolio |      |      |      |      |      |      |      |      |       |
|-----------------------------|-----------|------|------|------|------|------|------|------|------|-------|
|                             | 1         | 2    | 3    | 4    | 5    | 6    | 7    | 8    | 9    | 10    |
| Excess return (%)           |           |      |      |      |      |      |      |      |      |       |
| IVS <sub>ATM</sub>          | 0.97      | 1.08 | 0.97 | 0.98 | 0.82 | 0.80 | 0.77 | 0.69 | 0.57 | -0.03 |
| IVS <sub>OTM</sub>          | 1.01      | 1.02 | 0.91 | 0.78 | 0.82 | 0.76 | 0.67 | 0.74 | 0.59 | 0.30  |
| $\Delta$ IVS <sub>ATM</sub> | 0.85      | 0.96 | 0.90 | 0.86 | 0.85 | 0.82 | 0.82 | 0.64 | 0.61 | 0.28  |
| $\Delta$ IVS <sub>OTM</sub> | 0.86      | 0.91 | 0.79 | 0.85 | 0.85 | 0.85 | 0.79 | 0.69 | 0.60 | 0.40  |
| Skew                        | 0.95      | 1.03 | 0.97 | 0.84 | 0.79 | 0.77 | 0.81 | 0.64 | 0.52 | 0.29  |
| AVAR                        | 0.90      | 0.91 | 0.86 | 0.86 | 0.91 | 0.79 | 0.90 | 0.86 | 0.61 | 0.07  |
| Left-jump intensity         |           |      |      |      |      |      |      |      |      |       |
| IVS <sub>ATM</sub>          | 0.77      | 0.82 | 0.81 | 0.80 | 0.79 | 0.80 | 0.82 | 0.85 | 0.89 | 1.03  |
| IVS <sub>OTM</sub>          | 0.68      | 0.72 | 0.73 | 0.76 | 0.79 | 0.82 | 0.85 | 0.90 | 0.97 | 1.16  |
| $\Delta$ IVS <sub>ATM</sub> | 0.84      | 0.84 | 0.83 | 0.81 | 0.80 | 0.80 | 0.81 | 0.84 | 0.87 | 0.92  |
| $\Delta$ IVS <sub>OTM</sub> | 0.84      | 0.83 | 0.81 | 0.79 | 0.78 | 0.79 | 0.81 | 0.84 | 0.89 | 0.97  |
| Skew                        | 0.71      | 0.74 | 0.74 | 0.76 | 0.79 | 0.82 | 0.85 | 0.89 | 0.95 | 1.13  |
| AVAR                        | 0.79      | 0.79 | 0.77 | 0.76 | 0.77 | 0.79 | 0.82 | 0.86 | 0.92 | 1.08  |

- High IV spread have low return  $\Rightarrow$  IV spread identify change in future return ditribution
- Positive return mainly generated by long leg  $\Rightarrow$  Long buyer possess information
- High downside jump risk lower future return  $\Rightarrow$  Contrary to risk compensation theory

# Alphas Against Factor Models (DCBS=1)

|               | IVS <sub>ATM</sub> | IVS <sub>OTM</sub> | $\Delta$ IVS <sub>ATM</sub> | $\Delta$ IVS <sub>OTM</sub> | Skew               | AVAR               |
|---------------|--------------------|--------------------|-----------------------------|-----------------------------|--------------------|--------------------|
| (Intercept)   | -0.08<br>(0.10)    | -0.35***<br>(0.11) | -0.08<br>(0.13)             | -0.24*<br>(0.14)            | -0.12<br>(0.11)    | 0.01<br>(0.10)     |
| PC1           | 0.04<br>(0.06)     | 0.13*<br>(0.07)    | -0.15<br>(0.09)             | -0.12<br>(0.09)             | 0.12<br>(0.08)     | -0.10<br>(0.08)    |
| PC2           | -0.02<br>(0.05)    | -0.21***<br>(0.05) | 0.16***<br>(0.05)           | 0.08<br>(0.05)              | -0.14**<br>(0.06)  | 0.09*<br>(0.05)    |
| PC3           | -0.02<br>(0.03)    | -0.00<br>(0.03)    | -0.03<br>(0.03)             | -0.06*<br>(0.03)            | -0.07**<br>(0.03)  | -0.11***<br>(0.03) |
| PC4           | 0.01<br>(0.03)     | -0.02<br>(0.03)    | 0.05<br>(0.04)              | 0.06*<br>(0.03)             | -0.00<br>(0.03)    | 0.07<br>(0.05)     |
| PC5           | 0.04*<br>(0.02)    | 0.08***<br>(0.03)  | -0.07**<br>(0.03)           | -0.03<br>(0.03)             | 0.04<br>(0.03)     | -0.06*<br>(0.03)   |
| Leverage      | -0.57***<br>(0.06) | -0.30***<br>(0.06) | -0.41***<br>(0.05)          | -0.12***<br>(0.05)          | -0.33***<br>(0.06) | -0.30***<br>(0.06) |
| Vol. of. vol. | 0.07<br>(0.05)     | 0.09<br>(0.06)     | 0.10*<br>(0.05)             | 0.02<br>(0.04)              | 0.06<br>(0.05)     | -0.03<br>(0.07)    |
| ATM vol.      | -0.06<br>(0.06)    | -0.14***<br>(0.05) | -0.07<br>(0.06)             | -0.01<br>(0.05)             | -0.16***<br>(0.06) | -0.08<br>(0.07)    |
| LJI           | -0.06<br>(0.04)    | 0.19***<br>(0.05)  | -0.08<br>(0.06)             | -0.05<br>(0.04)             | 0.28***<br>(0.05)  | 0.26***<br>(0.07)  |
| Downside beta | 0.03<br>(0.03)     | -0.01<br>(0.04)    | 0.07<br>(0.04)              | 0.01<br>(0.03)              | 0.00<br>(0.03)     | 0.04<br>(0.04)     |
| Tail beta     | -0.06*<br>(0.03)   | 0.01<br>(0.04)     | 0.07**<br>(0.03)            | 0.00<br>(0.04)              | 0.08**<br>(0.04)   | 0.18***<br>(0.05)  |

- Stocks with low short-sell-cost loss significant alpha  $\Rightarrow$  IV spread return come from mispricing caused by limit to arbitrage

# Discussion

- Limited understanding of information source
  - Where information about future returns come from?
  - Institutional investor use option to express private information in advance.
  - IV spread Reflect fundamental information in advance.
- Feature Engineering Limitation
  - Complete option market is implied volatility surface in 3D space  $IV = f(K, T)$ .
  - Compress to 29 indicators only capture partial information.
  - CNN or Transformer to extract information from implied volatility surface.

Table 1

|                    | Mean (%) | Standard deviation (%) | Sharpe ratio | Low Pf. (%) | High Pf. (%) | Skewness | Kurtosis | Maximum drawdown |
|--------------------|----------|------------------------|--------------|-------------|--------------|----------|----------|------------------|
| CIV                | -5.10    | 37.59                  | -0.14        | 8.47        | 3.37         | 0.54     | 3.27     | 84.88            |
| PIV                | -8.81    | 37.29                  | -0.24        | 9.13        | 0.32         | 0.59     | 3.39     | 76.30            |
| $\Delta$ CIV       | 4.86     | 12.24                  | 0.40         | 3.81        | 8.67         | 0.58     | 12.49    | 35.94            |
| $\Delta$ PIV       | -2.38    | 12.09                  | -0.20        | 7.51        | 5.13         | -0.20    | 17.18    | 52.51            |
| O/S                | -6.97    | 15.46                  | -0.45        | 11.51       | 4.54         | 0.70     | 10.06    | 61.18            |
| VRP                | -0.16    | 13.00                  | -0.01        | 6.15        | 5.99         | -0.59    | 3.60     | 40.62            |
| $IVS_{ATM}$        | -11.95   | 10.73                  | -1.11        | 11.61       | -0.35        | -2.11    | 25.32    | 26.82            |
| $IVS_{OTM}$        | -8.53    | 12.57                  | -0.68        | 12.14       | 3.61         | -1.97    | 16.75    | 34.42            |
| $\Delta IVS_{ATM}$ | -6.77    | 7.53                   | -0.90        | 10.15       | 3.38         | 0.07     | 5.12     | 17.50            |
| $\Delta IVS_{OTM}$ | -5.63    | 6.82                   | -0.83        | 10.38       | 4.75         | -1.18    | 6.71     | 13.16            |
| Skew               | -7.95    | 10.19                  | -0.78        | 11.43       | 3.48         | -0.89    | 5.34     | 19.20            |
| $\beta_{Jump}$     | -0.82    | 15.51                  | -0.05        | 5.48        | 4.66         | 0.28     | 64.15    |                  |
| $\beta_{Vol}$      | -0.02    | 13.43                  | 0.00         | 5.48        | 5.47         | -1.24    | 5.84     | 71.76            |
| $\beta_{VIX}$      | -6.29    | 17.79                  | -0.35        | 8.53        | 2.24         | -1.76    | 12.03    | 42.29            |
| $\beta_{Skew}$     | -7.38    | 16.67                  | -0.44        | 9.26        | 1.88         | -0.49    | 5.66     | 51.76            |
| $\beta_{VRP}$      | -0.65    | 18.53                  | -0.04        | 6.41        | 5.76         | -0.51    | 7.96     | 73.85            |
| $\beta_{VSOtm}$    | -5.54    | 16.56                  | -0.33        | 8.23        | 2.69         | -0.40    | 5.02     | 62.09            |
| $\beta_{TSCall}$   | -3.53    | 16.96                  | -0.21        | 7.19        | 3.67         | -1.44    | 8.10     | 45.87            |
| $\beta_{TSPut}$    | -4.12    | 16.51                  | -0.25        | 7.74        | 3.63         | -1.54    | 10.34    | 51.10            |
| AVAR               | -10.03   | 13.93                  | -0.72        | 10.83       | 0.81         | -0.16    | 7.49     | 41.68            |
| VAR <sup>+</sup>   | -5.51    | 36.60                  | -0.15        | 8.72        | 3.21         | 0.60     | 4.26     | 81.31            |
| VAR <sup>-</sup>   | -8.98    | 37.15                  | -0.24        | 9.81        | 0.83         | 0.48     | 3.42     | 78.23            |
| VAR <sup>Q</sup>   | -7.22    | 37.25                  | -0.19        | 9.36        | 2.14         | 0.50     | 3.61     | 77.87            |
| KT                 | -6.70    | 37.31                  | -0.18        | 9.35        | 2.65         | 0.55     | 3.82     | 80.55            |
| Vol. of vol.       | -5.25    | 9.47                   | -0.55        | 11.15       | 5.91         | 0.06     | 3.55     | 24.17            |
| TSCALL             | -5.62    | 14.83                  | -0.38        | 8.61        | 2.99         | -0.19    | 12.07    | 21.91            |
| TSPUT              | -5.25    | 14.91                  | -0.35        | 7.92        | 2.66         | 0.18     | 10.08    | 29.90            |
| $\Delta$ TSCALL    | 0.64     | 9.78                   | 0.07         | 6.33        | 6.97         | -2.02    | 18.31    | 34.59            |
| $\Delta$ TSPUT     | -2.73    | 9.88                   | -0.28        | 7.61        | 4.87         | -1.83    | 18.47    | 43.46            |

| varlabel     | m        | s        | sharpe   | low      | high     | skewness | kurtosis | drawdown min | max      |          |
|--------------|----------|----------|----------|----------|----------|----------|----------|--------------|----------|----------|
| CIV          | -5.10348 | 37.59137 | -0.13576 | 8.471632 | 3.36815  | 0.537643 | 3.33798  | 84.87519     | -41.2127 | 46.84246 |
| PIV          | -8.81102 | 37.29341 | -0.23626 | 9.126987 | 0.315963 | 0.590509 | 3.463031 | 76.29666     | -39.4142 | 49.32936 |
| DeltaCIV     | 4.864581 | 12.23889 | 0.397469 | 3.809379 | 8.67396  | 0.582952 | 12.71032 | 35.93704     | -20.1457 | 22.52809 |
| DeltaPIV     | -2.37864 | 12.08795 | -0.19678 | 7.507259 | 5.128623 | -0.20474 | 17.47978 | 52.51087     | -26.2521 | 22.08172 |
| O/S          | -6.96615 | 15.46245 | -0.45052 | 11.51098 | 4.544827 | 0.703985 | 10.24633 | 61.18183     | -22.7839 | 30.061   |
| VRP          | -0.16135 | 13.00261 | -0.01241 | 6.150407 | 5.989061 | -0.59148 | 3.67951  | 40.61636     | -19.3351 | 12.50245 |
| IVS_ATM      | -11.9546 | 10.73496 | -1.11362 | 11.60707 | -0.34757 | -2.1198  | 25.75415 | 26.8164      | -29.9705 | 13.28264 |
| IVS_OTM      | -8.53002 | 12.57318 | -0.67843 | 12.14284 | 3.61282  | -1.9802  | 17.04632 | 34.42018     | -31.1684 | 13.90161 |
| DeltaIVS_ATM | -6.77448 | 7.532416 | -0.89938 | 10.15185 | 3.377362 | 0.065389 | 5.22359  | 17.50105     | -9.44463 | 12.56935 |
| DeltaIVS_OTM | -5.62573 | 6.81885  | -0.82503 | 10.37901 | 4.753281 | -1.18117 | 6.837711 | 13.15633     | -13.5595 | 6.162672 |
| Skew         | -7.94759 | 10.19044 | -0.77991 | 11.42534 | 3.477749 | -0.89177 | 5.44189  | 19.20081     | -18.9968 | 8.422382 |
| beta_Jump    | -0.81608 | 15.51435 | -0.0526  | 5.477595 | 4.661515 | 0.282208 | 3.052317 | 64.14529     | -18.4674 | 19.28262 |
| beta_Vol     | -0.01624 | 13.43435 | -0.00121 | 5.483266 | 5.467029 | -1.24855 | 5.953128 | 71.75719     | -21.3354 | 11.45695 |
| beta_VIX     | -6.28619 | 17.78835 | -0.35339 | 8.527712 | 2.241523 | -1.76816 | 12.24126 | 42.29375     | -35.3057 | 17.75035 |
| beta_Skew    | -7.38057 | 16.66506 | -0.44288 | 9.256515 | 1.875946 | -0.49449 | 5.766967 | 51.76316     | -27.3091 | 18.57121 |
| beta_VRP     | -0.65288 | 18.53076 | -0.03523 | 6.41253  | 5.759646 | -0.51063 | 8.105243 | 73.855       | -32.2707 | 27.61928 |
| beta_IVSOTM  | -5.54219 | 16.5553  | -0.33477 | 8.234153 | 2.691965 | -0.40466 | 5.120659 | 62.09495     | -25.5362 | 19.30371 |
| beta_TSCALL  | -3.52837 | 16.96033 | -0.20804 | 7.19492  | 3.666554 | -1.44976 | 8.249277 | 45.5425      | -29.8139 | 18.24275 |
| beta_TSPUT   | -4.11745 | 16.51391 | -0.24933 | 7.744629 | 3.627184 | -1.54777 | 10.52955 | 51.10266     | -34.4004 | 17.57002 |
| AVAR         | -10.027  | 13.92748 | -0.71994 | 10.83444 | 0.807462 | -0.16126 | 7.628888 | 41.67792     | -19.9767 | 22.89712 |
| VAR+         | -5.51298 | 36.60277 | -0.15062 | 8.723743 | 3.210765 | 0.607291 | 4.345292 | 81.31248     | -40.2992 | 53.36621 |
| VAR-         | -8.98151 | 37.14791 | -0.24178 | 9.814627 | 0.833114 | 0.485265 | 3.493237 | 78.22571     | -40.278  | 46.05475 |
| VARQ         | -7.21706 | 37.24986 | -0.19375 | 9.361341 | 2.144283 | 0.501167 | 3.685941 | 77.87404     | -41.3961 | 48.56004 |
| KT           | -6.69785 | 37.30683 | -0.17953 | 9.350526 | 2.652678 | 0.55211  | 3.903081 | 80.54543     | -41.1474 | 50.21813 |
| Vol. of vol. | -5.24512 | 9.469178 | -0.55391 | 11.15186 | 5.906741 | 0.062035 | 3.629356 | 24.17402     | -10.8788 | 13.87869 |
| TSCALL       | -5.62173 | 14.82884 | -0.37911 | 8.608154 | 2.986427 | -0.19373 | 12.28797 | 21.90677     | -25.2445 | 21.90677 |
| TSPUT        | -5.25303 | 14.91029 | -0.35231 | 7.916555 | 2.663522 | 0.177089 | 10.26817 | 29.89813     | -24.2827 | 20.12621 |
| DeltaTSCALL  | 0.644412 | 9.784595 | 0.06586  | 6.326888 | 6.971299 | -2.02709 | 18.62706 | 34.59213     | -20.1086 | 14.14267 |
| DeltaTSPUT   | -2.73336 | 9.87966  | -0.27667 | 7.606849 | 4.873491 | -1.84282 | 18.78765 | 43.46217     | -22.613  | 14.10003 |

Table 2

|             | CIV                | PIV                | $\Delta CIV$       | $\Delta PIV$      | O/S                | VRP             | $IVS_{ATM}$        | $IVS_{OTM}$        | $\Delta IVS_{ATM}$ | $\Delta IVS_{OTM}$ |
|-------------|--------------------|--------------------|--------------------|-------------------|--------------------|-----------------|--------------------|--------------------|--------------------|--------------------|
| (Intercept) | -0.20<br>(0.28)    | -0.49*<br>(0.28)   | 0.36*<br>(0.21)    | -0.19<br>(0.21)   | -0.56***<br>(0.16) | 0.06<br>(0.22)  | -1.18***<br>(0.18) | -0.85***<br>(0.19) | -0.63***<br>(0.14) | -0.53***<br>(0.14) |
| mktrf       | 0.63***<br>(0.08)  | 0.61***<br>(0.09)  | 0.09<br>(0.06)     | 0.06<br>(0.05)    | 0.18***<br>(0.05)  | -0.11<br>(0.07) | 0.04<br>(0.07)     | -0.07<br>(0.07)    | -0.02<br>(0.04)    | -0.01<br>(0.04)    |
| smb         | 1.09***<br>(0.11)  | 1.06***<br>(0.10)  | 0.17<br>(0.11)     | 0.11<br>(0.10)    | 0.00<br>(0.10)     | 0.10<br>(0.12)  | -0.03<br>(0.07)    | -0.17*<br>(0.09)   | 0.02<br>(0.05)     | 0.04<br>(0.05)     |
| hml         | -0.66***<br>(0.11) | -0.66***<br>(0.11) | 0.04<br>(0.07)     | -0.00<br>(0.07)   | -0.50***<br>(0.07) | 0.07<br>(0.11)  | 0.19*<br>(0.10)    | 0.34***<br>(0.09)  | -0.08<br>(0.05)    | -0.03<br>(0.05)    |
| rmw         | -1.44***<br>(0.16) | -1.46***<br>(0.14) | 0.04<br>(0.14)     | 0.05<br>(0.11)    | -0.45***<br>(0.14) | -0.02<br>(0.16) | 0.11<br>(0.10)     | 0.34***<br>(0.12)  | 0.18**<br>(0.08)   | 0.10*<br>(0.06)    |
| cma         | -0.41**<br>(0.19)  | -0.39*<br>(0.21)   | -0.28***<br>(0.10) | -0.23**<br>(0.09) | -0.21<br>(0.14)    | -0.08<br>(0.19) | 0.18<br>(0.17)     | -0.03<br>(0.14)    | 0.11<br>(0.09)     | 0.01<br>(0.08)     |
| umd         | -0.64***<br>(0.11) | -0.63***<br>(0.11) | -0.06<br>(0.06)    | -0.11*<br>(0.07)  | 0.27***<br>(0.08)  | 0.03<br>(0.11)  | 0.21**<br>(0.10)   | 0.22***<br>(0.07)  | -0.02<br>(0.04)    | 0.07*<br>(0.04)    |

|                     | Skew              | $\beta_{Jump}$     | $\beta_{Vol.}$  | $\beta_{VIX}$      | $\beta_{Skew}$   | $\beta_{VRP}$     | $\beta_{IVSOTM}$ | $\beta_{TSCALL}$   | $\beta_{TSPUT}$   | AVAR               |
|---------------------|-------------------|--------------------|-----------------|--------------------|------------------|-------------------|------------------|--------------------|-------------------|--------------------|
| (Intercept)         | -0.72**<br>(0.15) | 0.09<br>(0.23)     | 0.11<br>(0.23)  | -0.73***<br>(0.16) | -0.43<br>(0.28)  | -0.14<br>(0.34)   | -0.29<br>(0.27)  | -0.48*<br>(0.30)   | -0.45<br>(0.30)   | -0.63***<br>(0.19) |
| mktrf               | -0.06<br>(0.06)   | 0.13<br>(0.08)     | -0.10<br>(0.11) | 0.07<br>(0.07)     | -0.12*<br>(0.06) | -0.22**<br>(0.08) | -0.11<br>(0.07)  | 0.08<br>(0.07)     | 0.03<br>(0.07)    | 0.14**<br>(0.07)   |
| smb                 | -0.13<br>(0.09)   | -0.04<br>(0.13)    | -0.11<br>(0.13) | -0.33**<br>(0.14)  | -0.15<br>(0.11)  | -0.08<br>(0.13)   | -0.09<br>(0.10)  | -0.42***<br>(0.15) | -0.28**<br>(0.14) | 0.02<br>(0.11)     |
| hml                 | 0.20**<br>(0.09)  | -0.01<br>(0.13)    | -0.07<br>(0.14) | 0.18***<br>(0.06)  | -0.08<br>(0.15)  | 0.21<br>(0.13)    | -0.08<br>(0.13)  | 0.16<br>(0.11)     | 0.07<br>(0.10)    | -0.24**<br>(0.10)  |
| rmw                 | 0.18<br>(0.13)    | -0.58***<br>(0.18) | -0.02<br>(0.20) | 0.40*<br>(0.21)    | -0.02<br>(0.18)  | 0.42**<br>(0.18)  | 0.00<br>(0.16)   | 0.38*<br>(0.22)    | 0.30<br>(0.21)    | -0.67***<br>(0.14) |
| cma                 | -0.01<br>(0.13)   | 0.21<br>(0.18)     | -0.29<br>(0.23) | -0.07<br>(0.09)    | -0.10<br>(0.18)  | -0.03<br>(0.19)   | -0.08<br>(0.17)  | 0.02<br>(0.16)     | 0.09<br>(0.13)    | -0.10<br>(0.13)    |
| umd                 | 0.14***<br>(0.04) | -0.21**<br>(0.09)  | 0.14<br>(0.10)  | 0.16<br>(0.10)     | -0.11<br>(0.10)  | 0.27**<br>(0.12)  | -0.14<br>(0.09)  | 0.10<br>(0.13)     | 0.00<br>(0.11)    | -0.06<br>(0.06)    |
| Adj. R <sup>2</sup> | 0.21              | 0.23               | 0.09            | 0.15               | 0.01             | 0.24              | 0.01             | 0.18               | 0.09              | 0.47               |
| Num. obs.           | 311               | 308                | 308             | 311                | 311              | 311               | 311              | 311                | 311               | 311                |

| term      | CIV               | PIV               | DeltaCIV         | DeltaPIV        | O/S               | VRP            | $IVS_{ATM}$       | $IVS_{OTM}$       | $IVS_{ATM}$       | $IVS_{OTM}$       |
|-----------|-------------------|-------------------|------------------|-----------------|-------------------|----------------|-------------------|-------------------|-------------------|-------------------|
| const     | -0.2<br>-0.27     | -0.49*<br>-0.27   | 0.36*<br>-0.21   | -0.19<br>-0.21  | -0.56***<br>-0.15 | 0.06<br>-0.2   | -1.18***<br>-0.19 | -0.85***<br>-0.19 | -0.63***<br>-0.14 | -0.53***<br>-0.14 |
| mktrf     | 0.63***<br>-0.08  | 0.61***<br>-0.09  | 0.09<br>-0.06    | 0.06<br>-0.05   | 0.18***<br>-0.06  | -0.11<br>-0.07 | 0.04<br>-0.07     | -0.07<br>-0.07    | -0.02<br>-0.04    | -0.01<br>-0.04    |
| smb       | 1.09***<br>-0.1   | 1.06***<br>-0.09  | 0.17<br>-0.1     | 0.11<br>-0.11   | 0<br>-0.1         | 0.1<br>-0.11   | -0.03<br>-0.06    | -0.17*<br>-0.09   | 0.02<br>-0.05     | 0.04<br>-0.04     |
| hml       | -0.66***<br>-0.1  | -0.66***<br>-0.11 | 0.04<br>-0.08    | 0<br>-0.07      | -0.50***<br>-0.07 | 0.07<br>-0.11  | 0.19*<br>-0.1     | 0.34***<br>-0.09  | -0.08<br>-0.05    | -0.03<br>-0.05    |
| rmw       | -1.44***<br>-0.16 | -1.46***<br>-0.14 | 0.04<br>-0.13    | 0.05<br>-0.11   | -0.45***<br>-0.13 | -0.02<br>-0.16 | 0.11<br>-0.12     | 0.34***<br>-0.08  | 0.18**<br>-0.05   | 0.10*<br>-0.05    |
| cma       | -0.41**<br>-0.2   | -0.39*<br>-0.22   | -0.28**<br>-0.11 | -0.23**<br>-0.1 | -0.21<br>-0.16    | -0.08<br>-0.18 | 0.18<br>-0.16     | -0.03<br>-0.14    | 0.11<br>-0.09     | 0.01<br>-0.09     |
| umd       | -0.64***<br>-0.11 | -0.63***<br>-0.11 | -0.06<br>-0.06   | -0.11<br>-0.07  | 0.27***<br>-0.08  | 0.03<br>-0.11  | 0.21**<br>-0.08   | 0.22***<br>-0.04  | -0.02<br>-0.04    | 0.07**<br>-0.04   |
| Adj. R2   | 0.83              | 0.83              | 0.07             | 0.06            | 0.58              | 0              | 0.18              | 0.37              | 0.04              | 0.04              |
| Num. obs. | 311               | 311               | 310              | 310             | 311               | 311            | 311               | 311               | 310               | 310               |

| term      | Skew              | beta_Jump         | beta_Vol.      | beta_VIX          | beta_Skew      | beta_VRP          | beta_IVSO      | beta_TSCA         | beta_TSPU        | AVAR              |
|-----------|-------------------|-------------------|----------------|-------------------|----------------|-------------------|----------------|-------------------|------------------|-------------------|
| const     | -0.72***<br>-0.15 | 0.09<br>-0.24     | 0.11<br>-0.25  | -0.73***<br>-0.26 | -0.43<br>-0.34 | -0.14<br>-0.27    | -0.29<br>-0.34 | -0.48*<br>-0.27   | -0.45<br>-0.29   | -0.63***<br>-0.17 |
| mktrf     | -0.06<br>-0.05    | 0.13<br>-0.08     | -0.1<br>-0.11  | 0.07<br>-0.08     | -0.12<br>-0.07 | -0.22***<br>-0.07 | -0.11<br>-0.08 | 0.08<br>-0.07     | 0.03<br>-0.07    | 0.14**<br>-0.06   |
| smb       | -0.13<br>-0.08    | -0.04<br>-0.12    | -0.11<br>-0.13 | -0.33*<br>-0.18   | -0.15<br>-0.12 | -0.08<br>-0.12    | -0.09<br>-0.11 | -0.42***<br>-0.15 | -0.28**<br>-0.14 | 0.02<br>-0.11     |
| hml       | 0.20**<br>-0.09   | -0.01<br>-0.13    | -0.07<br>-0.14 | 0.18<br>-0.12     | -0.08<br>-0.15 | 0.21<br>-0.13     | -0.08<br>-0.13 | 0.16<br>-0.11     | 0.07<br>-0.12    | -0.24**<br>-0.1   |
| rmw       | 0.18<br>-0.12     | -0.58***<br>-0.17 | -0.02<br>-0.2  | 0.40*<br>-0.22    | -0.02<br>-0.19 | 0.42**<br>-0.17   | 0<br>-0.18     | 0.38*<br>-0.22    | 0.3<br>-0.21     | -0.67***<br>-0.14 |
| cma       | -0.01<br>-0.13    | 0.21<br>-0.18     | -0.29<br>-0.22 | -0.07<br>-0.19    | -0.03<br>-0.2  | -0.08<br>-0.18    | 0.02<br>-0.17  | 0.02<br>-0.15     | 0.09<br>-0.15    | -0.1<br>-0.15     |
| umd       | 0.14***<br>-0.05  | -0.21**<br>-0.08  | 0.14<br>-0.09  | 0.16<br>-0.14     | -0.11<br>-0.1  | 0.27**<br>-0.13   | -0.14<br>-0.09 | 0.1<br>-0.13      | 0<br>-0.12       | -0.06<br>-0.06    |
| Adj. R2   | 0.21              | 0.23              | 0.09           | 0.15              | 0.01           | 0.24              | 0.01           | 0.18              | 0.09             | 0.47              |
| Num. obs. | 311               | 308               | 308            | 311               | 311            | 311               | 311            | 311               | 311              | 311               |

Table 3

|                  | CAPM     | FF3      | FF4      | FF5      | FF7a     | FF7b     | HXZ      | HXZM     | PC5      | PC7      |
|------------------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|
| CIV              | -1.54*** | -1.49*** | -1.03*** | -0.51    | -0.07    | -0.26    | -0.24    | -0.22    | -0.44*   | 0.49**   |
| PIV              | -1.83*** | -1.78*** | -1.33*** | -0.79**  | -0.36    | -0.54**  | -0.53    | -0.51    | -0.78*** | 0.11     |
| ACIV             | 0.27     | 0.27     | 0.32     | 0.33*    | 0.35*    | 0.28*    | 0.34     | 0.35     | 0.47**   | 0.24     |
| APIV             | -0.30*   | -0.30*   | -0.23    | -0.25    | -0.21    | -0.26    | -0.21    | -0.19    | -0.11    | -0.40    |
| O/S              | -0.77*** | -0.68*** | -0.84*** | -0.43*** | -0.51*** | -0.54*** | -0.50*   | -0.56*** | -0.28*   | 0.08     |
| VRP              | 0.05     | 0.05     | 0.03     | 0.07     | 0.05     | 0.02     | 0.12     | 0.11     | 0.11     | -0.05    |
| IVS.ATM          | -0.92*** | -0.94*** | -1.08*** | -1.08*** | -1.18*** | -1.15*** | -1.19*** | -1.21*** | -1.43*** | -1.10*** |
| IVS.OTM          | -0.51**  | -0.54*** | -0.69*** | -0.75*** | -0.88*** | -0.81*** | -0.96*** | -0.96*** | -0.94*** | -0.96*** |
| DeltaIVS_ATM     | -0.52*** | -0.52*** | -0.51*** | -0.64*** | -0.63*** | -0.60*** | -0.56*** | -0.56*** | -0.63*** | -0.62*** |
| DeltaIVS_OTM     | -0.43*** | -0.43*** | -0.48*** | -0.50*** | -0.53*** | -0.51*** | -0.49*** | -0.50*** | -0.53*** | -0.51*** |
| Skew             | -0.53*** | -0.54*** | -0.63*** | -0.66*** | -0.75*** | -0.67*** | -0.82*** | -0.82*** | -0.77*** | -0.83*** |
| beta_Jump        | -0.29    | -0.29    | -0.14    | -0.01    | 0.12     | 0.07     | 0.10     | 0.11     | -0.23    | 0.24     |
| beta_Vol         | 0.09     | 0.12     | 0.03     | 0.18     | 0.12     | 0.12     | 0.06     | 0.04     | 0.41*    | 0.01     |
| beta_VIX         | -0.42    | -0.43*   | -0.54**  | -0.65*** | -0.73*** | -0.77*** | -0.77*** | -0.77*** | -0.43    | -0.34    |
| beta_Skew        | -0.56*   | -0.54*   | -0.47    | -0.48    | -0.43    | -0.48    | -0.47    | -0.46    | -0.24    | -0.15    |
| beta_VRP         | 0.27     | 0.26     | 0.07     | -0.01    | -0.20    | -0.17    | -0.16    | -0.18    | -0.01    | -0.34    |
| beta_IVSOTM      | -0.42    | -0.41    | -0.31    | -0.36    | -0.29    | -0.35    | -0.35    | -0.34    | -0.11    | 0.01     |
| beta_TSCALL      | -0.20    | -0.21    | -0.28    | -0.43*   | -0.48*   | -0.52*   | -0.55**  | -0.55**  | -0.17    | -0.10    |
| beta_TSPUT       | -0.26    | -0.27    | -0.28    | -0.45*   | -0.45    | -0.50*   | -0.51*   | -0.50*   | -0.23    | -0.12    |
| AVAR             | -1.10*** | -1.06*** | -1.00*** | -0.66*** | -0.60*** | -0.59*** | -0.72*** | -0.73*** | -0.74*** | -0.35*   |
| VAR+             | -1.50*** | -1.46*** | -1.00*** | -0.45    | -0.03    | -0.20    | -0.21    | -0.18    | -0.43**  | 0.30*    |
| VAR-             | -1.82*** | -1.77*** | -1.34*** | -0.77**  | -0.36    | -0.53**  | -0.52    | -0.51    | -0.78*** | 0.14     |
| VAR <sup>Q</sup> | -1.68*** | -1.63*** | -1.18*** | -0.63**  | -0.21    | -0.39    | -0.39    | -0.37    | -0.63*** | 0.21     |
| KT               | -1.64*** | -1.59*** | -1.13*** | -0.59*   | -0.16    | -0.34    | -0.33    | -0.31    | -0.58**  | 0.26     |
| Vol. of. vol.    | -0.40**  | -0.42**  | -0.31*   | -0.28    | -0.23    | -0.23    | -0.30*   | -0.32*   | -0.46**  |          |
| TSCALL           | -0.71*** | -0.67*** | -0.50*** | -0.27    | -0.14    | -0.19    | -0.28    | -0.26    | -0.02    | 0.18     |
| TSPUT            | -0.71*** | -0.67*** | -0.49*** | -0.31*   | -0.16    | -0.21    | -0.31*   | -0.29    | -0.05    | 0.00     |
| DeltaTSCALL      | 0.00     | 0.00     | 0.03     | 0.06     | 0.07     | 0.05     | 0.03     | 0.04     | 0.05     | -0.11    |
| DeltaTSPUT       | -0.26    | -0.25    | -0.25    | -0.27    | -0.27    | -0.28    | -0.25    | -0.25    | -0.21    | -0.44*   |

| varlabel      | capm     | ff3      | ff3_umd  | ff5      | ff5_umd_ic | ff5_umd_liq | q_umd    | PC5      | PC7               |
|---------------|----------|----------|----------|----------|------------|-------------|----------|----------|-------------------|
| CIV           | -1.54*** | -1.49*** | -1.03*** | -0.51    | -0.07      | -0.26       | -0.24    | -0.22    | -0.44* 0.49**     |
| PIV           | -1.83*** | -1.78*** | -1.33*** | -0.79**  | -0.36      | -0.54**     | -0.53    | -0.51    | -0.78*** 0.11     |
| DeltaCIV      | 0.27     | 0.27     | 0.32     | 0.33*    | 0.35*      | 0.28*       | 0.34     | 0.35     | 0.47** 0.24       |
| DeltaPIV      | -0.30*   | -0.30*   | -0.23    | -0.25    | -0.21      | -0.26       | -0.21    | -0.19    | -0.11 -0.4        |
| O/S           | -0.77*** | -0.68*** | -0.84*** | -0.43**  | -0.51***   | -0.54***    | -0.50*   | -0.56*** | -0.28* 0.08       |
| VRP           | 0.05     | 0.05     | 0.03     | 0.07     | 0.05       | 0.02        | 0.12     | 0.11     | 0.11 -0.05        |
| IVS_ATM       | -0.92*** | -0.94*** | -1.08*** | -1.08*** | -1.18***   | -1.15***    | -1.19*** | -1.21*** | -1.43*** -1.10*** |
| IVS_OTM       | -0.51**  | -0.54*** | -0.69*** | -0.75*** | -0.88***   | -0.81***    | -0.96*** | -0.96*** | -0.94*** -0.96*** |
| DeltaIVS_ATM  | -0.52*** | -0.52*** | -0.51*** | -0.64*** | -0.63***   | -0.60***    | -0.56*** | -0.56*** | -0.63*** -0.62*** |
| DeltaIVS_OTM  | -0.43*** | -0.43*** | -0.48*** | -0.50*** | -0.53***   | -0.51***    | -0.49*** | -0.50*** | -0.53*** -0.51*** |
| Skew          | -0.53*** | -0.54*** | -0.63*** | -0.66*** | -0.75***   | -0.67***    | -0.82*** | -0.82*** | -0.77*** -0.83*** |
| beta_Jump     | -0.29    | -0.29    | -0.14    | -0.01    | 0.12       | 0.07        | 0.1      | 0.11     | -0.23 0.24        |
| beta_Vol      | 0.09     | 0.12     | 0.03     | 0.18     | 0.12       | 0.12        | 0.06     | 0.04     | 0.41* 0.01        |
| beta_VIX      | -0.42*   | -0.43*   | -0.54**  | -0.65*** | -0.73***   | -0.77***    | -0.77*** | -0.77*** | -0.43 -0.34       |
| beta_Skew     | -0.56*   | -0.54*   | -0.47    | -0.48    | -0.43      | -0.48       | -0.47    | -0.46    | -0.24 -0.15       |
| beta_VRP      | 0.27     | 0.26     | 0.07     | -0.01    | -0.2       | -0.17       | -0.16    | -0.18    | -0.01 -0.34       |
| beta_IVSOTM   | -0.42    | -0.41    | -0.31    | -0.36    | -0.29      | -0.35       | -0.35    | -0.34    | -0.11 0.01        |
| beta_TSCALL   | -0.2     | -0.21    | -0.28    | -0.43*   | -0.48*     | -0.52*      | -0.55**  | -0.55*   | -0.17 -0.1        |
| beta_TSPUT    | -0.26    | -0.27    | -0.28    | -0.45    | -0.45      | -0.50*      | -0.51*   | -0.50*   | -0.23 -0.12       |
| AVAR          | -1.10*** | -1.06*** | -1.00*** | -0.66*** | -0.60***   | -0.59***    | -0.72*** | -0.73*** | -0.74*** -0.35**  |
| VAR+          | -1.50*** | -1.46*** | -1.00*** | -0.45    | -0.03      | -0.2        | -0.21    | -0.18    | -0.43** 0.30*     |
| VAR-          | -1.82*** | -1.77*** | -1.34*** | -0.77**  | -0.36      | -0.53**     | -0.52    | -0.51    | -0.78*** 0.14     |
| VARQ          | -1.68*** | -1.63*** | -1.18*** | -0.63**  | -0.21      | -0.39       | -0.39    | -0.37    | -0.63*** 0.21     |
| KT            | -1.64*** | -1.59*** | -1.13*** | -0.59*   | -0.16      | -0.34       | -0.33    | -0.31    | -0.58** 0.26      |
| Vol. of. vol. | -0.40**  | -0.42**  | -0.31*   | -0.28*   | -0.23      | -0.23       | -0.32*   | -0.30*   | -0.32* -0.46**    |
| TSCALL        | -0.71*** | -0.67*** | -0.50*** | -0.27    | -0.14      | -0.19       | -0.28    | -0.26    | -0.02 0.18        |
| TSPUT         | -0.71*** | -0.67*** | -0.49*** | -0.31*   | -0.16      | -0.21       | -0.31    | -0.29    | -0.05 0           |
| DeltaTSCALL   | 0        | 0        | 0.03     | 0.06     | 0.07       | 0.05        | 0.03     | 0.04     | 0.05 -0.11        |
| DeltaTSPUT    | -0.26    | -0.25    | -0.25    | -0.27    | -0.27      | -0.28       | -0.25    | -0.25    | -0.21 -0.44*      |

Table 4

| varlabel      | N      | beg_date | dash_col | end_date | BRT     | selected | selected_chars                                                                          |  |  |  |  |  |  |
|---------------|--------|----------|----------|----------|---------|----------|-----------------------------------------------------------------------------------------|--|--|--|--|--|--|
| CIV           | 756589 | Feb-96   | -        | Dec-21   | 8/151   | N        | mispricing_perf, ret_6_1, div12m_me, seas_1_1na, prc_highprc_252d, netdebt_me, sale_n   |  |  |  |  |  |  |
| PIV           | 756589 | Feb-96   | -        | Dec-21   | 1/151   | Y        | mispricing_perf, ret_6_1, div12m_me, seas_1_1na, prc_highprc_252d, netdebt_me, sale_n   |  |  |  |  |  |  |
| DeltaCIV      | 743725 | Mar-96   | -        | Dec-21   | 30/151  | Y        | mispricing_perf, ret_6_1, seas_1_1na, prc_highprc_252d, netdebt_me, op_at, gp_at, misp  |  |  |  |  |  |  |
| DeltaPIV      | 743725 | Mar-96   | -        | Dec-21   | 27/151  | N        | mispricing_perf, ret_6_1, ret_9_1, seas_1_1na, prc_highprc_252d, netdebt_me, op_at, gp_ |  |  |  |  |  |  |
| O/S           | 713187 | Feb-96   | -        | Dec-21   | 7/151   | Y        | mispricing_perf, ret_6_1, ret_9_1, div12m_me, seas_1_1na, prc_highprc_252d, netdebt_m   |  |  |  |  |  |  |
| VRP           | 755277 | Feb-96   | -        | Dec-21   | 69/151  | N        | mispricing_perf, ret_6_1, div12m_me, seas_1_1na, prc_highprc_252d, netdebt_me, sale_n   |  |  |  |  |  |  |
| IVS_ATM       | 755990 | Feb-96   | -        | Dec-21   | 4/151   | Y        | mispricing_perf, ret_6_1, div12m_me, seas_1_1na, prc_highprc_252d, netdebt_me, op_at    |  |  |  |  |  |  |
| IVS_OTM       | 755990 | Feb-96   | -        | Dec-21   | 11/151  | Y        | mispricing_perf, ret_6_1, div12m_me, seas_1_1na, prc_highprc_252d, netdebt_me, sale_n   |  |  |  |  |  |  |
| DeltaIVS_ATM  | 744016 | Mar-96   | -        | Dec-21   | 34/151  | Y        | mispricing_perf, ret_6_1, seas_1_1na, prc_highprc_252d, netdebt_me, op_at, gp_at, misp  |  |  |  |  |  |  |
| DeltaIVS_OTM  | 744016 | Mar-96   | -        | Dec-21   | 44/151  | Y        | mispricing_perf, ret_6_1, div12m_me, seas_1_1na, prc_highprc_252d, netdebt_me, op_at    |  |  |  |  |  |  |
| Skew          | 756589 | Feb-96   | -        | Dec-21   | 12/151  | Y        | mispricing_perf, ret_6_1, div12m_me, seas_1_1na, prc_highprc_252d, netdebt_me, sale_n   |  |  |  |  |  |  |
| beta_Jump     | 748502 | May-96   | -        | Dec-21   | 110/151 | N        | mispricing_perf, ret_6_1, div12m_me, seas_1_1na, prc_highprc_252d, netdebt_me, sale_n   |  |  |  |  |  |  |
| beta_Vol      | 748502 | May-96   | -        | Dec-21   | 64/151  | N        | mispricing_perf, ret_6_1, div12m_me, seas_1_1na, prc_highprc_252d, netdebt_me, sale_n   |  |  |  |  |  |  |
| beta_VIX      | 755982 | Feb-96   | -        | Dec-21   | 5/151   | Y        | mispricing_perf, ret_6_1, div12m_me, seas_1_1na, prc_highprc_252d, netdebt_me, op_at    |  |  |  |  |  |  |
| beta_Skew     | 755982 | Feb-96   | -        | Dec-21   | 14/151  | Y        | mispricing_perf, ret_6_1, div12m_me, seas_1_1na, prc_highprc_252d, netdebt_me, op_at    |  |  |  |  |  |  |
| beta_VRP      | 755982 | Feb-96   | -        | Dec-21   | 90/151  | N        | mispricing_perf, ret_6_1, div12m_me, seas_1_1na, prc_highprc_252d, netdebt_me, sale_n   |  |  |  |  |  |  |
| beta_IVSOTM   | 755982 | Feb-96   | -        | Dec-21   | 17/151  | Y        | mispricing_perf, ret_6_1, div12m_me, seas_1_1na, prc_highprc_252d, netdebt_me, sale_n   |  |  |  |  |  |  |
| beta_TSCALL   | 755982 | Feb-96   | -        | Dec-21   | 34/151  | Y        | mispricing_perf, ret_6_1, div12m_me, seas_1_1na, prc_highprc_252d, netdebt_me, op_at    |  |  |  |  |  |  |
| beta_TSPUT    | 755982 | Feb-96   | -        | Dec-21   | 16/151  | Y        | mispricing_perf, ret_6_1, div12m_me, seas_1_1na, prc_highprc_252d, netdebt_me, sale_n   |  |  |  |  |  |  |
| AVAR          | 755277 | Feb-96   | -        | Dec-21   | 10/151  | Y        | mispricing_perf, ret_6_1, div12m_me, seas_1_1na, prc_highprc_252d, netdebt_me, sale_n   |  |  |  |  |  |  |
| VAR+          | 755277 | Feb-96   | -        | Dec-21   | 10/151  | N        | mispricing_perf, ret_6_1, div12m_me, seas_1_1na, prc_highprc_252d, netdebt_me, sale_n   |  |  |  |  |  |  |
| VAR-          | 755277 | Feb-96   | -        | Dec-21   | 5/151   | Y        | mispricing_perf, ret_6_1, div12m_me, seas_1_1na, prc_highprc_252d, netdebt_me, sale_n   |  |  |  |  |  |  |
| VARQ          | 755277 | Feb-96   | -        | Dec-21   | 9/151   | N        | mispricing_perf, ret_6_1, div12m_me, seas_1_1na, prc_highprc_252d, netdebt_me, sale_n   |  |  |  |  |  |  |
| KT            | 755277 | Feb-96   | -        | Dec-21   | 9/151   | N        | mispricing_perf, ret_6_1, div12m_me, seas_1_1na, prc_highprc_252d, netdebt_me, sale_n   |  |  |  |  |  |  |
| Vol. of. vol. | 756146 | Feb-96   | -        | Dec-21   | 21/151  | N        | mispricing_perf, ret_6_1, div12m_me, seas_1_1na, prc_highprc_252d, netdebt_me, sale_n   |  |  |  |  |  |  |
| TSCALL        | 755992 | Feb-96   | -        | Dec-21   | 8/151   | Y        | mispricing_perf, ret_6_1, div12m_me, seas_1_1na, prc_highprc_252d, netdebt_me, sale_n   |  |  |  |  |  |  |
| TSPUT         | 755992 | Feb-96   | -        | Dec-21   | 8/151   | Y        | mispricing_perf, ret_6_1, div12m_me, seas_1_1na, prc_highprc_252d, netdebt_me, op_at    |  |  |  |  |  |  |
| DeltaTSCALL   | 747773 | Mar-96   | -        | Dec-21   | 85/151  | N        | mispricing_perf, ret_6_1, div12m_me, seas_1_1na, prc_highprc_252d, netdebt_me, sale_n   |  |  |  |  |  |  |
| DeltaTSPUT    | 747773 | Mar-96   | -        | Dec-21   | 31/151  | Y        | mispricing_perf, ret_6_1, div12m_me, seas_1_1na, prc_highprc_252d, netdebt_me, op_at    |  |  |  |  |  |  |